

Analysis First-Year Exam, January 2025, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f : [0, 1] \rightarrow \mathbb{R}$ be Riemann integrable. If f is continuous at 0, does it follow that

$$\int_0^1 f(x^n) dx \rightarrow f(0) \quad \text{as } n \rightarrow \infty?$$

If so, give a proof; if no, give a counterexample.

2. Let $(f_n)_{n=0}^\infty$ be a sequence of continuous real-valued functions on the metric space X , in which $(x_n)_{n=0}^\infty$ is a convergent sequence.

(i) Prove that if $(f_n)_{n=0}^\infty$ is uniformly convergent then $(f_n(x_n))_{n=0}^\infty$ converges.

(ii) Show by example that $(f_n(x_n))_{n=0}^\infty$ can fail to converge if $(f_n)_{n=0}^\infty$ is merely pointwise convergent.

3. Let Φ be an equicontinuous family of real-valued functions on the connected metric space X . Prove that if Φ is bounded at one point of X then Φ is bounded at every point of X . [Note: to say that Φ is bounded at $p \in X$ is to say that $\{f(p) : f \in \Phi\} \subseteq \mathbb{R}$ is bounded.]

4. Let $(f_n)_{n=0}^\infty$ be a sequence of measurable real-valued functions on some measure space. Prove that each of the following sets is measurable.

(i) $A = \{\omega : f_n(\omega) \rightarrow 0 \text{ as } n \rightarrow \infty\}$.

(ii) $B = \{\omega : \sum_n f_n(\omega) \text{ converges absolutely}\}$.

5. This problem has two parts.

(i) State the Monotone Convergence Theorem.

(ii) On a measure space, let $(f_n)_{n=0}^\infty$ be a sequence of non-negative measurable functions converging pointwise to the function f . Prove that if there exists a real number K such that each of the integrals $\int f_n d\mu$ is at most K , then f is integrable and $\int f d\mu \leq K$.